

Academia Nuts

“I hate you Steve Taylor.” “Why don’t you die Steve Taylor.”

The words come unbidden to my lips like a reflex. Morning noon and night. First thing when I wake up, last thing when I go to sleep. When I am on the toilet. After I make love.

Clearly I have a screw loose. Steve Taylor drove me crazy.

Stephen Taylor was a junior assistant professor at the California Institute of Technology computer science department in the mid 1990s. He was a sociopath. I was one of his graduate students which meant that I was completely under his control.

Steve spoke with a lower-class English accent. A small man, he loved to command large supercomputers.

When I first started working with him he asked me what my goal was post-PhD. I said I wanted to be a professor at Stanford. He said “all right we will get you a position at Stanford then.” I was foolish enough to believe him. This motivated me for quite a while until I realized it was nonsense.

Once Steve asked me “can you explain the concept of Eigenvalue?” My heart sunk into my boots. This was a basic sophomore level mathematical concept from linear algebra. If he didn’t understand Eigensystems he had no hope of succeeding in his chosen field of computational fluid dynamics.

We were at a funding opportunity at Rockwell Defense Systems in Hawthorne. Another team from UCLA was there working on finite element models of electromagnetic structures. This was something we didn’t know anything about. Steve took a superior attitude, aloof and disinterested. He was careful to ask no questions. After all they were merely a public university and we were the great Caltech.

We had successfully committed fraud. We misrepresented our abilities in order to gain funding,

On the way back Steve was jubilant. “Getting government money is easy! I am a great man, a man of vision. They don’t have people like us at these defense contractors. They need real experts!”

I wear my hair short to disguise the fact that I smoke pot. It is not approved of at Caltech although I know many people who smoke it including professors. Steve also wears his hair in a military style and my ruse worked. He tuned the radio to right wing hate monger Rush Limbaugh. “Rush is right” he said to me.

I change the subject by asking about his wife who has been doing poorly since they lost their house in foreclosure. A childlike creature, he married her apparently because her

parents had money. They were two of a kind, both overweight, dressed the same, wore their hair the same, asexual. Together they looked like Tweedledum and Tweedledee.

He blamed her for the foreclosure saying that she was responsible for paying the bills while he worked. There was some truth to that, he worked like a madman and was rarely at home. This caused her distress. He once said to me that she was a liability to his career because she didn't understand his ambition.

Steve's work all concerned simulations of weapons of war. Rocket launch vehicles, detonations, everything that involved killing people made him happy. He thought his work was more important than fundamental sciences like biology and chemistry.

There was no music or poetry in Steve's work. It was all nose to the grindstone, painfully won progress. No pain, no gain. He wanted his students to suffer like he did. He didn't understand my attitude. For me intellectual work was stimulating and liberating. It expanded my soul and made me happy. No pain needed.

Herbert Keller was professor of applied mathematics at Caltech and one of its thought leaders. Herb's research was on numerical bifurcation analysis of differential equations. Steve was doing a research project under Herb as part of his assistant professorship. Unfortunately Steve didn't have the necessary mathematical background and was floundering. He had recruited me to work with him because I was an expert in bifurcation theory and scientific supercomputing. He hoped that I could save his project and his career.

The first time I met Herb in his office I saw a beautiful golden light in the space between us. I was used to mystical experiences so I understood this was an auspicious sign. And sure enough over time Herb became one of my best friends and strongest supporters at Caltech.

The project was to do numerical bifurcation analysis of the Navier-Stokes equations of fluid mechanics in the Taylor-Couette experiment. Bifurcation analysis meant determining points on the solution manifold at which multiple solutions exist. This marks the transition between qualitatively different solutions. The subject goes back to Poincare in the 1800s. By traversing these different solutions we could trace the route from simple laminar flow to full chaotic turbulence.

The problem I was faced with was that the solver for the pressure equation ran forever without completing. The solver had been written by a postdoc who had left without completing it. Herb called in one of his senior students Andrew Conley. Andrew look at the solver for five minutes and declared "David Haraar is incompetent." It seemed that David had used a solution method, red-black Gauss-Seidel iteration, that does not converge on finite difference discretizations of the Navier-Stokes equations. So all the money that had been spent on the project was wasted, and Herb was upset because he did not have a paper to present at the semi-annual Taylor-Couette symposium.

Andrew was a friendly and helpful person and we soon got to talking about my research. I was looking to find a parallel algorithm to solve the an optimization problem in task scheduling for parallel supercomputers. I was looking at diffusion as an algorithmic metaphor but I was stuck. I didn't know the eigenvalues of the diffusion equation. I asked

Andrew to help me and he obliged. He knew the eigenvalues by heart. “How did you know that?” I asked him. “Just numerical analysis seasoning” was his response.

Andrew was graduating with a Ph.D. and he didn’t want to leave Caltech. Steve had an opening for a postdoc. I proposed to Steve that we hire Andrew who was an expert in computational fluid dynamics. Steve was open to the idea and I encouraged Andrew to pursue it.

I told Steve that Andrew was very helpful to me in reviewing my work. “Good. Get everything you can from this guy” was Steve’s response. At first I thought that Steve was testing Andrew, but eventually I came to see a sinister motive.

Steve scheduled a working session at his home to produce a paper about my diffusion work. His wife was deferential as she left us to work in the living room. Steve didn’t write any text but instead shuffled my notes around. He told me to write up the result and submit it to the Journal of Parallel and Distributed Computing.

I did what he said and the reviews came back from the journal. They were scathing and sarcastic. The paper was soundly rejected. The editor wrote “I think there is something here and I invite you to rewrite the paper and resubmit it.” Steve put his nose in the air and said “don’t change a thing, resubmit the paper as is.”

Steve thought he could bluff or bully his way into getting published. He didn’t understand that he was merely a junior assistant professor and nobody was scared of him. The subject editor was a well known computer scientist John Gustafson who was an alumnus and friend of the Caltech department.

Steve’s instructions were impossible to obey so I took three days and rewrote the paper from scratch. This time it had a narrative and conclusion and proved the algorithm was correct and efficient. I submitted the revised paper to the journal. This time the reviews were more positive but it was still not accepted for publication. Steve didn’t say anything at the time but eventually told me that I was stealing from him by writing the paper.

After a couple of weeks Steve told me “I am not going to consider your friend Andrew. I want you to tell him.” I asked what was wrong. “Look at what he did!” Steve handed me a paper that Andrew wrote. The work was clearly inspired by my work on diffusion, but just as clearly was Andrew’s original ideas.

I felt a little bit betrayed that Andrew had not shown me the work he was doing. But I didn’t mind if he learned from me, it was only fair. I didn’t believe in intellectual materialism.

“Andrew is stealing from you! I want you to take this paper and put it in the DARPA report with your name on it. After all you deserve the credit for it.” Steve had a crafty look on his face like he knew he was doing something wrong. He grimaced. “There are no ethics in academia.”

I was shocked. Andrew didn’t steal anything from me. He was just inspired by my work. It goes against the entire academic ethos to steal a paper in this way. And it amounts to fraud to include it in a DARPA report. I didn’t know what to say.

Needless to say I didn't carry out his instructions. Instead I reported the incident to a senior faculty member in the computer science department. His words were "The only safe policy with Steve Taylor is to stay as far away from him as possible." He said this was a serious matter and he would take it up with the department. Thanks a lot, no help.

Steve was angry that I did not follow his orders to steal Andrew's paper. He gave me a written order to produce a thesis by a certain date. It was the first time I learned of his manic behavior and emotional reactivity. The implication was that if I did not complete the work on time I could be dismissed.

Jan van de Snepscheut was the graduate option representative for the computer science department. He had been recruited from Europe with high expectations. His graduate advisor was the legendary Edsger Dijkstra. Dijkstra had written in his recommendation that Jan would be one of the greatest computer scientists.

Jan was considered a genius by the department and the institute. The only person who had more influence was Carver Mead, the legendary VLSI inventor. Jan specialized in mind bending puzzles. Here is an example of a puzzle from his collection.

A circle of light bulbs

n light bulbs are arranged in a circle. Each light bulb is either on or off. In the middle of the circle is a push button. Pushing the button has the following effect on the light bulbs:

a light bulb is on after the button is released just when this light bulb and its right-hand neighbor were both on or both off before the button was pressed.

Show that, if n is a power of 2, all light bulbs are on after the button is pushed n times, independent of the initial configuration.

By the way, the property that all n light bulbs are on after pushing the button n times, independently of the initial configuration, is not only implied by but equivalent to the property that n is a power of 2.

Here is a solution due to Rick Hehner. We prove the property by induction on n . The induction base is that in a circle of 1 light bulb, the bulb is its own right neighbor, it is the same as itself, so 1 push of the button causes it to be on.

For the induction step, consider a circle of $2n$ lights. That's a circle of n pairs of lights. Consider a pair of lights to be on if and only if both lights in the pair are on. Now check that a pair of button pushes has the desired property:

Suppose a pair and its right neighbor pair are both on (ie all 4 lights on). Then after 1 push at least the first 3 lights are on, and after 2 pushes at east the first 2 lights are on, so the left pair is on. The other three cases can be checked similarly.

So, by the induction hypothesis, after n pairs of pushes, the n pairs of lights are all on. In other words, after $2n$ pushes, all $2n$ lights are on.

Jan was a popular teacher and took a personal interest in students, especially the most talented ones. He was elitist and had no use for Steve Taylor.

Jan was at that time the graduate option representative. Jan approached me in the hall. "You have a very hard deadline to meet." I replied "Yes but we are very soft people." However I couldn't really take the threat of the deadline seriously. It was such a change from the way Steve had treated me before,

I wrote Steve a long email accounting for the work I had been doing. He had suggested that I write a survey of computational fluid dynamics algorithms and I reminded him that this took a lot of my time. I wrote to him about the work I had been doing in learning more about bifurcation theory and the numerical experiments I had conducted. And I described the work on the diffusion algorithm. Before I sent it off to him I cc'd it to Jan. One minute later I got a reply from Steve: "I don't care what you think you are working on." Steve didn't realize the message and his reply had been cc'd to Jan.

I completed the work on the diffusion algorithm and handed it in to Steve. Crisis averted for now. But the nature of our working relationship had been fundamentally changed.

I asked Steve for feedback on my work. He said "Mathematicians like you are a dime a dozen. I hire and fire you like that!" He snapped his fingers and walked away.

This was disheartening. I didn't understand why he was treating me this way. I had done very good work for Steve on the Taylor-Couette problem and had earned praise from Herb Keller. I worked hard on the diffusion problem. Also I am only an amateur mathematician, my three degrees are all in computation.

I sent the revised paper to Jan and wrote "perhaps it could be published." He replied the next day "your work is original and creative. Of course it should be published!"

I learned long ago that administrators are the real powers in the department. They can make or break you if they want to. I made friends with a warm motherly administrator who will go nameless. She took an interest in all of the students. I brought my two year old daughter in to campus to visit with her. My daughter was pre-verbal. She praised my daughter for being extremely alert and watching everything that happened. She had been that way since the first day we brought her home from the hospital.

She confided in me that Steve had tried to get rid of her. Without thinking I said "you know with him that is partially a racial thing." She said "I know I've been facing that all of my life." I had never heard Steve say anything overtly racist but I believed it because it was consistent with rest of his world view.

Today she told me that Steve and Jan were having a royal battle and that it was "totally out of control". I told her that Steve had mentioned in his group meeting that he was having a conflict with Jan and that none of us were to speak to Jan. Of course I ignored that directive.

The administrator told me that Jan reported Steve said "if you interfere with my student again I will burn your fucking house down!" She said "I think he was talking about you. The

senior faculty are having a secret meeting off campus where they called in the big gun Carver Mead. Jan has proposed terminating Steve's employment contract for reasons of mental illness."

She also told me that Jan was now the department chair and had identified me to the institute as an outstanding student. "This is very good for you" she said. Apparently they track the students in this way. Of course neither Jan nor Steve told me anything about this.

I had reached a point where I came to believe that everything Steve told me was a deception. In retrospect this was probably going too far but he had lied and deceived me so many times I couldn't feel otherwise. Needless to say this did not help the professor-student relationship.

The administrator went on to say that Steve was openly talking about firing me, that I "was not worth it." This made me feel bitterly. I had invested years in my graduate education. I had tried to do the right thing and had gotten in trouble with Steve because he was crooked. I was in good standing with the senior members of the department.

A few days later a message came to me through the administrator from Jan. He requested that I meet with him in his office. I had never seen a formal request of this sort so of course I complied.

I went to see Jan in his oak paneled office on the second floor of the Jorgensen building. I closed the door behind me. Jan asked "why did you stop working with your former advisor? Steve told me that you had a personality conflict."

I said "Steve has told me many things that proved to be complete ... fabrication!" I said the word fabrication with a twang of emotion.

Jan said "Yes I have had the same experience. So what was the reason?"

I said "I wrote a disgrace of a paper. It was very bad. I don't have a background in biology. It was so bad I could not work with that advisor any more."

Jan nodded and seemed to be thinking. I said "When I heard about the threat against you I thought to myself 'this is a mind that is cracking under pressure'."

Jan gasped and swiveled his chair around so that I could not see him. I heard the sounds of sobbing. He stayed this way for a long time. I did not know what to think. Eventually he turned the chair around to face me again. "Yes" is all he said, He seemed devastated.

Jan said "You are a good student and you deserve to complete your PhD. What is the resolution for all of this?"

I said "Could I become your student?"

Jan said "I think that would be advisable. It will take me a little time to arrange. Meanwhile please continue as though nothing has changed."

Steve started sending Jan a series of profane email messages. Jan complained to the institute provost. The provost described the messages as “pretty raw.”

I spent the next few days avoiding Steve and waiting for my liberation. I was a little bit afraid of Jan because I had imposter syndrome and was afraid that he would find flaws in me. But the situation with Steve was untenable.

A week passed. I came into campus in the morning and the halls were uncharacteristically abuzz with people. I saw a police car parked outside the building. There was a television news crew in the hallway. I entered Steve’s basement laboratory and knocked on the door. “What is going on around here?” I asked Steve.

Steve looked at me with absolutely no emotional expression. “Last night Jan died.”

I could feel the hairs stand up on the back of my neck.

“How did it happen?” I asked.

“Last night he burned his house down with him in it. It just shows this could happen to you at any time.”

I can hear the blood rushing in my ears. I can see that Steve is trying to stop from smiling. He must be insane. How does he think he can get away with this? Was the last sentence a threat or just a manner of speaking? I decided I had better interpret it as a threat because the risk of failing to do so was too high. My head started reeling.

I mumbled something incoherent and bolted from the room. Out in the basement hallway my vision was distorted. It was like a drug experience except that I don’t do drugs. Patches of color seemed to detach from the walls and were not binding into solid objects. I recognized that my visual cortex was malfunctioning, overwhelmed by what I had just heard.

I closed my eyes and the riot of color was replaced by blessed cool darkness. I felt my way like a blind man across the hallway to the stairs. Then I ascended the stairs one at a time with my eyes tightly shut. When I got to the top I could see again. The sky was a beautiful blue, birds were singing in the trees. A splendid southern California morning.

I made it to my car but my head was reeling. A couple of blocks from campus was a gun shop with a big cartoon drawing of Yosemite Sam firing pistols. Suddenly this seemed like the answer. I pulled over abruptly.

In the shop a 40-ish woman was behind the counter wearing a halter top. She leaned forward with her arms pushing her breasts up. She put down a cigarette and said with a smile “What can I do you for?”

“I think I need to buy a gun. A handgun,”

“There will be a two week waiting period on that. Are you all right with that?”

“I don’t think I can wait that long.” I said.

She said “Do you have a *situation*?”

“Yes I think so.”

She said “Dave get over here”. A gigantic mountain of a man detached himself from the background where he had been standing motionless and unnoticed. He looked like a biker and had arms as thick as trees. He could snap me in half with one hand,

“Dave give him one of your business cards.” The card said *NRA Regional Membership Manager*.

“Dave was bodyguard for Gordon Liddy a few years back. Let him know if you need anything.”

I thanked both of them and exited with the card in my pocket. I realized this was not the answer.

I drove home erratically to the townhouse that my wife and I share with our two young daughters. My wife asks why I am home early.

“There has been a big power struggle at Stanford. Jan van de Snepscheut has been killed. I am afraid that Steve Taylor may have murdered him. I don’t think we are safe here. We need to move as soon as possible.”

She is angry now. “Every Caltech professor you have introduced me to has been crazy. Why do you keep getting involved with these bad people? Why do we have to move?”

“Look I can’t help it. I have to work with who is available to me. I didn’t know Steve was crazy when I joined his lab.”

Somehow we moved house in 36 hours, 50 miles away. Somewhere Steve cannot find me. I watch the news impatiently waiting to hear that Steve has been arrested for murder. At night I was electrified with rage. I felt like part of my brain was melting, like wet concrete. I implanted in it a hatred for Steve Taylor. A deep hypnotic suggestion. The brain froze into that form and forever after I have had PTSD about Steve. Like hiccups, I cannot stop these curses.

“Steve you are an evil man. Why won’t you die.”

After a couple of days I haven’t seen any such news and I realized that I had misread the situation. I slip back onto campus to see my friend the administrator.

“It’s good to see you” she said. “I’m glad you are all right. Steve was looking for you. He asked everyone if they knew how to reach you.”

She told me that Jan had committed suicide. In the middle of the night he attacked his wife with an axe, set the house on fire, and tried to trap his children in the burning house. They all escaped. Jan’s body was found in a chair, dead of smoke inhalation, with a pile of his academic papers in his lap. She told me Jan was a recovered schizophrenic.

I was nauseated. This was not the Jan I thought I knew.

“Goddammit Steve! Yer an evil man!”

The administrator went on to tell me there was a female undergraduate in the department who had become impregnated and abandoned by one of the male graduate students. Jan got involved and fell in love with the undergraduate. They had an affair and this became public knowledge. Jan was humiliated by the exposure.

That sounded more like the Jan I thought I knew. He was a rescuer. Just as he tried to rescue me from Steve Taylor he tried to rescue the undergraduate.

“Steve, yer an evil man,”

The Pasadena police department was interested in the conflict between Steve and Jan. The day after the suicide a detective went through Jan’s file cabinets. There was an entire drawer devoted to Steve Taylor. The detective carefully photocopied every piece of paper in the drawer. But in the end they did not bring any charges. I think at least extreme emotional distress and depraved indifference to suffering were worthy of charges.

“I’ll burn yer fucking house down!”

I was without an advisor for six months. The senior faculty arranged several opportunities for me although some were unsuitable. I made a connection with professor Joel Franklin in the applied math department. I showed him my work on diffusion and he told me that his intuition was that I would be good at studying interior point methods. We spent an hour together and made a special connection that I recognized had mystical elements. At the end Joel said to me “you remind me tremendously of Donald Knuth when we was a student here.” High praise indeed.

“Steve I hate you. Yer an evil man!”

Unfortunately this relationship was scuttled by Eric van de Velde an instructor in Applied Math. He told Joel that my work on diffusion was unimportant. Rather than form his own opinion Joel accepted Eric’s judgement. After that we could no longer work together since diffusion was supposed to be the basis of my dissertation.

I was so angry during this time that I contacted author Michael Meloan and told him my story. In turn he wrote a compelling fictionalized version called “The Cutting Edge” that he published in WIRED magazine in March 1996.

“Steve yer a monster.”

I was depressed for six months and wore black every day. The stress level was so high that I lost the skin on the palms of my hands. I had to wear cotton gloves.

I wanted some attention so I wrote a letter to Edsger Dijkstra and included the revised paper. I pointed out that he had written the first paper predicting diffusion as a method for task placement in parallel computers. I asked if he had any advice for me.

Dijkstra wrote back to say that he did not understand the mathematics in my paper. This surprised me greatly because it was only linear systems analysis. This is something any electrical engineer learns as an undergraduate. If one of the gods of computer science does not know core engineering mathematics then what does that imply for the field?

“Steve you are a sadist!”

I also found that I had access to the editor backups Steve used to send email. I couldn't read his email directly but I found some choice items. For example I learned that he had safety schools in the event that he didn't get tenure at Caltech. These were the University of California Santa Barbara and the University of Utah.

One particularly juicy item was marked “Confidential Report to the Central Intelligence Agency.” In it I read that Steve was surveilling his colleagues in the computational fluid dynamics community. I wondered what they would have thought about this if they knew.

One day I found a correspondence with the head of DARPA funding Bob Lucas. Lucas wrote “you can send in any proposal you like, it won't get funded.” Sounds like Steve's DARPA report was a failure.

“I hate you Steve, I wish you would die.”

Steve published the diffusion work in secret at the 1995 International Conference on Parallel Processing. He claimed to be the inventor of the ideas which was false. I only found out about it because by good fortune the paper was awarded “outstanding paper of the year” and the conference contacted me. I had to attend the conference and have a showdown with Steve. When I told Joel Franklin about the award he became very angry with Eric.

“Steve Taylor you should die.”

I ended up completing my PhD with one of the other senior faculty and with a very nice man who was an expert in computer graphics. I made sure that he did not come in contact with any of the preceding psychodrama. Nonetheless he told me that Steve had come to see him and told him that I would steal all of his ideas. “He didn't seem sane to me” was his summary.

The Cutting Edge appeared in WIRED magazine the same month that Steve's tenure committee started meeting. It caused a minor sensation on the campus. I sent copies to Steve's safety schools.

“If you interfere with my student again I'll burn yer fucking house down!”

Steve went for 10 interviews at other schools and none of them made him an offer. Probably because his work had made no impact on the field outside of his own laboratory. He spent a year and a half hanging on at Caltech until he finally got an offer from Syracuse University. He had previously referred to Syracuse as a “backwater”. He spent three years there until they terminated his contract.

“Steve yer a fraud yer a phony.”

I attended Herb Keller's weekly research meetings faithfully for 6 years. As the sole computer scientist I didn't contribute a lot to the discussions but I learned a tremendous amount about solving differential equations.

One day Herb was lecturing about solving Magneto-Hydrodynamics equations. He made an off hand remark about matrix preconditioners that caused worlds to collide in my head.

Abruptly I had a sensation like summersaulting then I was out of my body looking down at the room from the ceiling. I saw all of the students in the room, Herb at the lectern, and over my head was a slowly rotating *sahasrar* chakra. This is a classical yogic meditation experience called the thousand petalled lotus. It is a sign of advanced enlightenment, it means the kundalini shakti has purified the lower chakras and is now working on the highest chakra at the crown of the head. The sahasrar chakra is considered the seat of enlightenment.

"Goddamit Steve Taylor yer sick."

I have had similar experiences in the presence of powerful gurus and mystics and now I realized that Herb himself was such a mystic. I knew that Herb was regarded as the spiritual father of the applied mathematics community and now I understood why. I would not have had this uplifting experience if it were not for Herb's own energy. He was in a high state of consciousness, the result of his life long work in mathematics.

"Steve Taylor yer an evil man."

When I completed my PhD the department chair invited me to apply for a tenure track position at Caltech. I was flattered but I demurred, I did not think my mental health was good enough. The last thing the world needed was another crazy Caltech professor. I needed a low pressure existence.

And so I have completed a relatively typical career in Silicon Valley. I invented something called the *scalable visualization array* with funding from the Department of Energy. Now that I am retired I have returned to research and am working with one of my dissertation committee members on methods to solve classical optimization problems using differential equations. So all is well that ends well.

"I hate Stephen Taylor."